Logarithms

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https://www.alphaclasses.com/

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https://sourceforge.net/u/rajatkalia/profile

https://www.youtube.com/@rajatkaliaeducation

https://www.lulu.com/spotlight/rajatkalia11

https://amazon.com/author/rajatkalia

https://rk-edu.blogspot.in/

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Chapter 1

Logarithms Worksheet

1.1 Logarithmic Differentiation(Revisiting Logarithms)

Let us first learn the basic definition of logarithms. First of all , we have a exponential equation of the form $a^{\alpha} = b$. This equation can be written in the logarithmic form as$\alpha = log_a b$. So, we understand that logarithms is just another way of writing an equation which has exponents involved in it.

Logarithms have few basic properties :

- $\log_a p = \dfrac{\log_b p}{\log_b a}$ (Base Change Formula)
- $\log_a pq = \log_a p + \log_a q$
  $\log_a p^n = n \log_a p$

Now suppose, we have a function of the form,
$y = f(x) = [u(x)]^{v(x)}$.
By taking logarithm (to base e) the above may be rewritten as
$\ln y = v(x) \ln[u(x)]$
Using chain rule we may dierentiate this to get
$\dfrac{1}{y}.\dfrac{dy}{dx} = v(x).\dfrac{1}{u(x)}.u'(x) + v'(x)\ln[u(x)]$
The main point to be noted in this method is that f (x) and u(x) must always be positive as otherwise their logarithms are not defined.

Q: Differentiate the following functions w.r.t. x

a) $f(x) = \sqrt{\dfrac{(x-3)(x^2+8)}{x^2+3x+4}}$
b) $f(x) = x^{\sin x}, x > 0$
c) $f(x) = cosx.cos2x.cos3x$
d) $f(x) = (\ln x)^{\cos x}$
e) $f(x) = (\ln x)^x + x^{\ln x}$
f) $f(x) = (\sin x)^x + \sin^{-1}\sqrt{x}$

Q1: Find the domains of definition of the following functions:

a)$\log\left(\dfrac{x^3\left(x^2-4\right)}{\left(x^2-1\right)(x+3)}\right)$
b)$\sqrt{\log x}$
c) $\sqrt{\log_x (2-x)}$

Q2 Matrix 1: Under Column I , some equations are given . Under Column II, some solutions satisfying some of the equations are given. Match the entry in Column I with the solution satisfying it in Column II.{Note: [ ] is the Greatest Integer Function}

Column I	**Column II**
(P) $\left\|\sin^{-1} x\right\| = \sin^{-1} x + \frac{\pi}{6}$	**(A)**-$\frac{1}{2}$
(Q) $\ln\|x\| = \|\ln x\|$	**(B)** 1
(R) $\|[x]\| = [\|x\|]$	**(C)** $\frac{3}{2}$
(S) $\left\|x^2 - 3x + 2\right\| > x^2 - 3x + 2$	**(D)** 2

Q3 Comprehension 1: A function $f_n(x)$ is defined for all $n \epsilon N$ and $f_{n+m}(x)$ is defined as $f_{n+m}(x) = f_n(f_m(x))$ where $f_1(x) = \frac{2x-1}{x+1}$ for $x \epsilon R - \{-1\}$. Using this definition , $f_2(x) = f_1(f_1(x)) = \frac{x-1}{x}$ for $x \epsilon R - \{-1, 0\}$. Similarly $f_3(x) = f_1(f_2(x)) = \frac{2-x}{1-2x}$ for $x \epsilon R - \left\{-1, 0, \frac{1}{2}\right\}$and so on. Based on the information, answer the questions below.

1: The domain of definition of $g(x) = |\ln(f_{34}(x))|$ is

a) $(-\infty, 1) - \left\{-1, 0, \frac{1}{2}\right\}$

b) $(-\infty, 1] - \left\{-1, 0, \frac{1}{2}\right\}$

c) $(1, \infty) - \{2\}$

d) None of these

2: The complete solution set of $|f_{71}(x)| > \left|\frac{1}{f_{73}(x)}\right|$ is

a) $(-1, 1)$

b) $\mathbb{R} - \{2\} - [-1, 1]$

c) $\mathbb{R} - \{-1, 1, 2\}$

d) None of these

3: The values of x for which $|f_{56}(x)| > f_{56}(x)$ belong to the interval

a) $\left(\frac{1}{2}, 2\right)$

b) $(2, \infty)$

c) $(0, 1)$

d) None of these

Q4: (i) Prove that for the logarithmic function $y = \ln|x|$, if the argument takes values in a G.P. , then the corresponding values of the function y are in A.P.

(ii)Prove that for the exponential function $y = e^x$, if the argument takes values in a A.P.. , then the corresponding values of the function y are in G.P.